

Week 4

1. Express $A = \begin{bmatrix} 0 & 5 \\ 1 & 3 \end{bmatrix}$ as the product of three elementary matrices.

$$A = \begin{bmatrix} 0 & 5 \\ 1 & 3 \end{bmatrix} \xrightarrow[E_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}]{R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & 3 \\ 0 & 5 \end{bmatrix} \xrightarrow[E_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1/5 \end{bmatrix}]{R_2/5} \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \xrightarrow[E_3 = \begin{bmatrix} 1 & -3 \\ 0 & 1 \end{bmatrix}]{R_1 - 3R_2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

$$E_1^{-1} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$E_2^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 5 \end{bmatrix}$$

$$E_3^{-1} = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$$

$$E_3 E_2 E_1 A = I \rightarrow A = E_1^{-1} E_2^{-1} E_3^{-1} I = E_1^{-1} E_2^{-1} E_3^{-1} \\ = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 0 & 5 \end{bmatrix} \cdot \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$$

$$[\text{check: } \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 0 & 5 \end{bmatrix} \cdot \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 & 3 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} 0 & 5 \\ 1 & 3 \end{bmatrix} = A \checkmark]$$

2. Let $A = \begin{bmatrix} -2 & 1 & 0 \\ -3 & 3 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 0 & 1 \\ 2 & -1 & 0 \end{bmatrix}$.

Find an invertible matrix U so that $B = UA$, and then express U as a product of elementary matrices.

B can be obtained from A by the following row operations:

$$A \xrightarrow[E_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}]{R_2 \leftrightarrow R_1} \begin{bmatrix} -3 & 3 & 1 \\ -2 & 1 & 0 \end{bmatrix} \xrightarrow[E_2 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}]{-R_2} \begin{bmatrix} -3 & 3 & 1 \\ 2 & -1 & 0 \end{bmatrix} \xrightarrow[E_3 = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}]{R_1 + 3R_2} \begin{bmatrix} 3 & 0 & 1 \\ 2 & -1 & 0 \end{bmatrix} = B$$

That is,

$$E_3 E_2 E_1 A = B, \text{ or in other words: } U = E_3 E_2 E_1$$

$$U = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} -3 & 1 \\ -1 & 0 \end{bmatrix}$$

product of elementary matrices.

$$[\text{check } U \text{ is invertible: } \det(U) = \det(E_3) \det(E_2) \det(E_1) = 1 \cdot (-1) \cdot (-1) = 1 \neq 0 \checkmark]$$

3. Find the following determinant by expanding along the second row and first column.

$$\text{row 2} \rightarrow \begin{vmatrix} 1 & 2 & 1 \\ 2 & 1 & 3 \\ 2 & 1 & 1 \end{vmatrix}$$

Laplace expansion over second row:

$$\begin{aligned} & \begin{vmatrix} 1 & 2 & 1 \\ 2 & 1 & 3 \\ 2 & 1 & 1 \end{vmatrix} = 2 \cdot (-1)^{(2,1)} \begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} + 1 \cdot (-1)^{(2,2)} \begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} + 3 \cdot (-1)^{(2,3)} \begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix} \\ & = 2 \cdot (-1) \cdot \begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} + 1 \cdot (1) \cdot \begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} + 3 \cdot (-1) \cdot \begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix} \\ & = -2 \cdot 1 + 1 \cdot (-1) - 3 \cdot (-3) = -2 - 1 + 9 = 6. \end{aligned}$$

similarly, for col 1: $1 \cdot (1 \cdot 1 - 3 \cdot 1) - 2(2 \cdot 1 - 1 \cdot 1) + 2(2 \cdot 3 - 1 \cdot 1) = -2 - 2 + 10 = 6$.

4. Find the determinant using row operations.

$$\begin{vmatrix} 1 & 2 & 1 \\ 2 & 3 & 2 \\ -4 & 1 & 2 \end{vmatrix}$$

$$\begin{vmatrix} 1 & 2 & 1 \\ 2 & 3 & 2 \\ -4 & 1 & 2 \end{vmatrix} \xrightarrow{R_2 - 2R_1, R_3 + 4R_1} \begin{vmatrix} 1 & 2 & 1 \\ 0 & -1 & 0 \\ 0 & 9 & 6 \end{vmatrix} \xrightarrow{R_3 + 9R_2} \begin{vmatrix} 1 & 2 & 1 \\ 0 & -1 & 0 \\ 0 & 0 & 6 \end{vmatrix} \xrightarrow{\text{upper-triangular}} 1 \cdot (-1) \cdot 6 = -6$$

5. Tell whether the statement is true or false.

(a) If A is a 3×3 matrix with a zero determinant, then one column must be a multiple of some other column. *False: example. $\det \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} = 0$*

(b) If any two columns of a square matrix are equal, then the determinant of the matrix equals zero. *True: Do an elementary column operation to get a zero col, then expand det over that col.*

(c) For A and B two $n \times n$ matrices, $\det(A+B) = \det(A) + \det(B)$. *False: example. Let $A=B=I_2$. $\det(A+B) = \det(2I_2) = 4$ but $\det(A) + \det(B) = 1+1=2$*

(d) For A an $n \times n$ matrix, $\det(3A) = 3 \det(A)$. *False. $\det(3A) = 3^n \det(A)$. e.g. $A=I_2$.*

(e) If A^{-1} exists then $\det(A^{-1}) = \det(A)^{-1}$. *True. $\det(A)^{-1} = \frac{1}{\det(A)} = \det(A^{-1})$*

(f) If B is obtained by multiplying a single row of A by 4 then $\det(B) = 4 \det(A)$. *True. expand over that row and compare.*

(g) For A an $n \times n$ matrix, $\det(-A) = (-1)^n \det(A)$. *True: $\det(kA) = k^n \det(A)$.*

(h) If A is a real $n \times n$ matrix, then $\det(A^T A) \geq 0$. *True. $\det(A^T A) = \det(A^T) \det(A) = \det(A) \cdot \det(A) = (\det(A))^2 \geq 0$*

(i) Cramers rule is useful for finding solutions to systems of linear equations in which there is an infinite set of solutions. *False, because in that case $\det(A) = 0$.*

6. Here is a matrix

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 2 & 1 \\ 3 & 1 & 1 \end{pmatrix}$$

Determine whether the matrix has an inverse by finding whether the determinant is non zero. If the determinant is nonzero, find the inverse using the formula for the inverse which involves the cofactor matrix.

$$\det(A) = 1 \cdot 1 + 3 \cdot 2 = 7 \neq 0 \Rightarrow \text{matrix has an inverse.}$$

$$A^{-1} = \frac{1}{7} \begin{pmatrix} (-1)^{1+1} \begin{vmatrix} 2 & 1 \\ 1 & 1 \end{vmatrix} & (-1)^{1+2} \begin{vmatrix} 0 & 1 \\ 3 & 1 \end{vmatrix} & (-1)^{1+3} \begin{vmatrix} 0 & 2 \\ 3 & 1 \end{vmatrix} \\ (-1)^{2+1} \begin{vmatrix} 1 & 0 \\ 1 & 1 \end{vmatrix} & (-1)^{2+2} \begin{vmatrix} 1 & 0 \\ 3 & 1 \end{vmatrix} & (-1)^{2+3} \begin{vmatrix} 1 & 2 \\ 3 & 1 \end{vmatrix} \\ (-1)^{3+1} \begin{vmatrix} 2 & 0 \\ 2 & 1 \end{vmatrix} & (-1)^{3+2} \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} & (-1)^{3+3} \begin{vmatrix} 1 & 2 \\ 0 & 2 \end{vmatrix} \end{pmatrix} = \frac{1}{7} \begin{pmatrix} 1 & 3 & -6 \\ -2 & 1 & 5 \\ 2 & -1 & 2 \end{pmatrix}^T$$

$$= \frac{1}{7} \begin{pmatrix} 1 & -2 & 2 \\ 3 & 1 & -1 \\ -6 & 5 & 2 \end{pmatrix}$$

$$\left[\text{check } AA^{-1} = I: A \left(\frac{1}{7} A^{-1} \right) = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 2 & 1 \\ 3 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -2 & 2 \\ 3 & 1 & -1 \\ -6 & 5 & 2 \end{pmatrix} = \begin{pmatrix} 7 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 7 \end{pmatrix} = 7I \checkmark \right]$$

Q: How do you know $A^{-1}A = I$?